## 🎓 HOW TO CREATE A GROUPED FREQUENCY DISTRIBUTION USING EXCEL (Pivot Table Method)

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### 1️⃣ Make sure your raw data are ready

Your data are already in the first column of the Excel table **Swim_times**.

Example:

| Swim_times |
|---|
| 84.7 |
| 88.3 |
| 91.0 |
| … |

No additional preparation is needed.

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### 2️⃣ Insert a Pivot Table

Click anywhere inside the column containing the swim times.

Go to:

**Insert → PivotTable**

Select:

- **New Worksheet**
- Click **OK**

Excel creates an empty PivotTable layout.

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### 3️⃣ Build the Pivot Table

Drag the variable **Swim_times**:

**➤ Into the Rows area**

Excel will display every individual value.

**➤ Into the Values area**

Excel should automatically use **Count**, which gives frequency (how many observations).

If not:

Right-click → **Value Field Settings** → **Count**

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### 4️⃣ Group the data into intervals (the key step)

Right-click any value in the PivotTable's **Row Labels** column → choose **Group...**

You will see:

- **Starting at: the lower real limit of the first group — 59.5 for the group 60–64, because the times are recorded to tenths (so that 84.7 s falls into 85–89, as in the textbook)**
- **Ending at**
- **By** (the class width)

Now choose the class width you want:

**✓ By = 5 (as used in the textbook)**

or

**✓ By = 6 (also completely correct — just produces fewer classes)**

📌 **Both choices are statistically correct** as long as:

- all intervals have equal width
- intervals cover the entire range
- there is no overlap
- the real limits are respected for continuous data (Starting at = 59.5, not 60)

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### 5️⃣ Excel generates the grouped frequency distribution

Example output:

| Interval | Count of Swim_times |
|---|---|
| 60–64 | 2 |
| 65–69 | 2 |
| 70–74 | 4 |
| ... | ... |

This is your grouped frequency table.

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### 6️⃣ (Optional) Create a Histogram directly from the Pivot Table

1. Select the Pivot Table
2. Go to **Insert → Column Chart**

3. Format axis labels if needed

You now have a clean histogram reflecting your chosen class width.

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### 🟦 Important Statistical Note for Students

This is the pedagogical core message:

**There is not only one "correct" class width.**
Both 5 and 6 (or any reasonable simple number such as 2, 5, 10) are fully acceptable,
**as long as intervals are equal, logical, and cover the full data range.**

The width of 5 gives more classes (more detail),
while the width of 6 gives fewer classes (simpler overview).

**Neither is wrong — the difference is in the level of detail, not in correctness.**

This mirrors the logic from Chapter 2 of the textbook.

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### 🟦 Ultra-Short Summary for Students

(Perfect as a slide or cheat sheet)

1. **Insert Pivot Table**
2. **Swim_times → Rows & Values (Count)**
3. **Right-click → Group → choose width (5 or 6)**
4. Pivot Table instantly creates grouped frequencies
5. (Optional) Add a chart for histogram

# References


Microsoft. (n.d.). *Group or ungroup data in a PivotTable*. Microsoft Support. Retrieved October 27, 2026, from https://support.microsoft.com/en-us/excel/get-started/group-or-ungroup-data-in-a-pivottable

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